

Problem

A system has the transfer function

$$H(s) = \frac{6s}{(s+1)(s+2)}$$

(a) Find the impulse response of the system.

(b) Determine the output $y(t)$, given that the input is $x(t) = u(t)$.

Step-by-step solution

Step 1 of 7

(a)

The system transfer function is,

$$H(s) = \frac{6s}{(s+1)(s+2)}$$

$$\frac{Y(s)}{X(s)} = \frac{6s}{(s+1)(s+2)} \dots\dots (1)$$

For an impulse,

$$x(t) = \delta(t)$$

The Laplace of impulse is,

$$X(s) = 1$$

Step 2 of 7

Recall equation (1).

$$\frac{Y(s)}{X(s)} = \frac{6s}{(s+1)(s+2)}$$

Substitute 1 for $X(s)$ in the above equation.

$$\frac{Y(s)}{1} = \frac{6s}{(s+1)(s+2)}$$

$$Y(s) = \frac{6s}{(s+1)(s+2)}$$

Apply partial fractions to simplify the above equation.

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2} \dots\dots (2)$$

Step 3 of 7

Calculate the constant A by using residue method.

$$\begin{aligned} A &= \left[(s+1)Y(s) \right]_{s=-1} \\ &= \left[(s+1) \frac{6s}{(s+1)(s+2)} \right]_{s=-1} \\ &= \frac{6(-1)}{-1+2} \\ &= -6 \end{aligned}$$

Calculate the constant B by using residue method.

$$\begin{aligned} B &= \left[(s+2)Y(s) \right]_{s=-2} \\ &= \left[(s+2) \frac{6s}{(s+1)(s+2)} \right]_{s=-2} \\ &= \frac{6(-2)}{-2+1} \\ &= 12 \end{aligned}$$

Step 4 of 7

Recall equation (2).

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2}$$

Substitute -6 for A and 12 for B in the above equation.

$$Y(s) = \frac{-6}{s+1} + \frac{12}{s+2}$$

Apply Inverse Laplace transform.

$$L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{-6}{s+1}\right\} + L^{-1}\left\{\frac{12}{s+2}\right\}$$

It is known that,

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

Therefore,

$$\begin{aligned} y(t) &= -6e^{-t}u(t) + 12e^{-2t}u(t) \\ &= [-6e^{-t} + 12e^{-2t}]u(t) \end{aligned}$$

Therefore, the impulse response, $y(t)$ of a system is $\boxed{[-6e^{-t} + 12e^{-2t}]u(t)}$.

Step 5 of 7

(b)

The input is,

$$x(t) = u(t)$$

The Laplace of impulse is,

$$X(s) = \frac{1}{s}$$

Recall equation (1).

$$Y(s) = \frac{6s}{(s+1)(s+2)} X(s)$$

Substitute $\frac{1}{s}$ for $X(s)$ in the above equation.

$$\begin{aligned} Y(s) &= \frac{6s}{(s+1)(s+2)} \left(\frac{1}{s} \right) \\ &= \frac{6}{(s+1)(s+2)} \end{aligned}$$

Apply partial fractions to simplify the above equation.

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2} \dots\dots (3)$$

Step 6 of 7

Calculate the constant A by using residue method.

$$\begin{aligned} A &= \left[(s+1)Y(s) \right]_{s=-1} \\ &= \left[(s+1) \frac{6}{(s+1)(s+2)} \right]_{s=-1} \\ &= \frac{6}{-1+2} \\ &= 6 \end{aligned}$$

Calculate the constant B by using residue method.

$$\begin{aligned} B &= \left[(s+2)Y(s) \right]_{s=-2} \\ &= \left[(s+2) \frac{6}{(s+1)(s+2)} \right]_{s=-2} \\ &= \frac{6}{-2+1} \\ &= -6 \end{aligned}$$

Step 7 of 7

Recall equation (3).

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2}$$

Substitute 6 for A and -6 for B in the above equation.

$$Y(s) = \frac{6}{s+1} - \frac{6}{s+2}$$

Apply Inverse Laplace transform.

$$L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{6}{s+1}\right\} - L^{-1}\left\{\frac{6}{s+2}\right\}$$

It is known that,

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

Therefore,

$$\begin{aligned} y(t) &= e^{-t}u(t) - 6e^{-2t}u(t) \\ &= [6e^{-t} - 6e^{-2t}]u(t) \end{aligned}$$

Therefore, the output, $y(t)$ of a system is $\boxed{[6e^{-t} - 6e^{-2t}]u(t)}$.